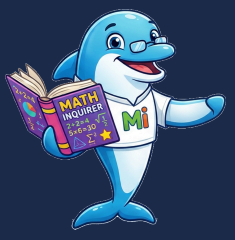
TUTOR GUIDE • LEVEL 1 • TERM 2 • LESSON 9 OF 10

Picture Equations



Big idea: A shape is a hidden number, and the same shape is always the same number. In a picture equation each shape is a number. The same shape is always the same number. We find the number, then check it in every line.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Addition and Subtraction; Multiplication and Division (reasoning about equality) (A number sentence with a missing number and "=" as the same amount on both sides (Primary 1, Addition and Subtraction, p. 31); Multiplication within 40 (Primary 1, Multiplication and Division, p. 31); A symbol that stands for an unknown number, and taking one line from another (letters for unknowns are Primary 6, p. 43)) Strand label for the tutor only. BEYOND PRIMARY 1 ON PURPOSE. One line with one hidden number (like 4 + box = 10) is Primary 1 work and fills Start here. Using the same shape in two lines, taking one line away from another, checking an answer in every line and a times line are not taught at school at this age (letters for unknowns are Primary 6, p. 43): this lesson uses drawn shapes and boxes only, never letters, always with counters or cups to hold the numbers. Think harder (two shapes, two lines) and Go further (three shapes, a times line, a sum and a product) go past the school Primary 1 syllabus. Owner decision, 22 Oct 2026: strong content with scaffolding, said plainly to families. Say this on the parent page and in the opening of the guide; do not describe the lesson as MOE-aligned or syllabus-aligned. Reading order for a times line: heart × circle is heart groups of circle; either order is accepted. Contest tag (tutor only): symbol equations and substitution; taking one line from another.
Objectives	• Say that a shape in a picture equation is a hidden number, and that the same shape is the same number in every line. • Find one hidden number in a line such as circle + square = 10 when the square is known, and in a line such as heart + heart = 14. • Check an answer by putting the number back into the line. • Think harder: use two lines about the same shapes (take one line from the other) and check the answer in both lines. • Go further: a times line, three shapes at once, and a sum and a product together.
Joining cold	**Remember warm-up covers:** A box in a number sentence hides a number: 6 + box = 10 (Term 1, taught again with a ten-frame); The same number twice: box + box = 12 (taught again with two equal plates); 4 groups of 5 as 4 × 5 (Term 2 Lesson 5, taught again with a picture). Only counting to 20 and adding small numbers. Nothing from an earlier lesson is needed: the warm-up shows the ten-frame, two equal plates and four bags with a picture, and the × sign is said again as groups of. The word shape stands for a number is taught in Explore. Fairly safe to miss, with one exception: Mi's Challenge Quest in Term 2 Lesson 10 has one picture-equation gate (a heart + heart total and a friend's number to check). A child who misses this lesson watches the recording and does workbook pages 1 and 2 before the quest. A child who joins here without Term 2 Lessons 5 and 6 can follow, because the warm-up shows groups of and the times sign again, and the times lines are in Think harder and Go further only.

Child materials	• The printed workbook with the ten-frames and counters kit page (page 6) and the shape cards kit page (page 7): a grown-up cuts out the 20 counters and the 15 shape cards beforehand • About 30 small things of one kind (counters, buttons, dry beans or pasta) in case more counters are needed • 2 small cups, mugs or bowls • The 15 shape cards from page 7 (three each of circle, square, triangle, heart and star). With no printer, draw the shapes on 15 small pieces of paper, or use up to 3 pieces of paper for each shape • A pencil, a rubber and spare paper
Tutor materials	• The slide deck • A set of paper shape cards (three each of circle, square, triangle, heart and star), a cup and some counters to show on camera if you wish • A visible clock No ruler is needed. Nothing needs to be bought: everything can be found at home. No matches, toothpicks or anything sharp. Small things never go near the mouth or ears; keep them away from younger brothers and sisters. Cups stay on the table.
Tiers	Start here (7 questions, pages 1 and 2), Think harder (3, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit pages (pages 6-7). Labels are neutral; nobody is told which page they are on.
Contest-style move	Used at Think harder, Go further and the Investigation: a symbol stands for one number in every line; take one line from another to find a shape; check the answer in every line; a sum and a product together. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first. Check that each shape in a puzzle is a different number (T1: 6 and 7; T2: 6 and 9; T3: 3 and 5; G1: 5, 7 and 11; G2: 8 and 6; G3: 5 and 4) so you can answer a child who asks.
- Tell parents a week before: use the workbook pages and both kit pages from the printed term book (send a PDF only if a page is missing, to print at 100%) (a grown-up cuts out the 20 counters and the 15 shape cards: three each of circle, square, triangle, heart and star), and find about 30 counters or buttons and two cups. Keep the cards in an envelope. Small things stay away from mouths and younger children. Next lesson is Mi's Challenge Quest: with this lesson's materials, send the Quest Log (in a closed envelope) and the grown-up note.
- Have the slides open. Have your own shape cards, a cup and counters to show on camera if you wish.
- Check the recording captures the tutor and the shared screen. Children's video is excluded; any voice that is not muted is recorded. Greet the whole group warmly, say the recording notice, then start the recording, and do not use children's first names while it runs.
- Be plain with families: one line with a hidden number is Primary 1 work, but using the same shape in two lines, taking one line from another and the times line are beyond the school Primary 1 syllabus on purpose. Do not call the lesson MOE-aligned. Never use letters for unknowns; shapes and boxes only. The counters are the method, not a rule to recite.
- Slides s-explore3, s-m2 and s-m3 keep the method off the screen while you ask: the steps appear on the slide after. Read every question with its totals and the four options of E2 aloud: the figures on the slides are small.

Safeguarding and recording (this lesson)

- Counters, cups and small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp. Keep small things away from younger brothers and sisters. Cups stay on the table so that nothing is spilled.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a board up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is not muted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Counting aloud is quiet; a child may stay seated.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Picture Equations	1:40
2–8	**Remember warm-up**	6	3: Warm-up: a box hides a number, The same number twice, 4 groups of 5	4:10
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:04
12–24	**Explore**	12	4: Explore: a shape and a pile, The same shape twice, Two lines, same shapes, Two lines, same shapes	9:44
24–36	**Mini-lesson**	12	9: Three rules, Find it, then check it, Find it, then check it, Take one line from another, Take one line from another, Check in both lines, A times line, Your turn, Check: your turn	9:57
36–40	**Brain break**	4	1: Clap and whisper	3:17
40–70	**Workbook**	30	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	24:58
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:50
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:53
82–85	**Look-back**	3	1: Look-back	2:23
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:57
	Total	**90**		**69:52**

The script needs 69:52 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 82:46. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 40 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 40, skip in this order: Start with: cut the times line slide (s-m4) in the Mini-lesson to one question; Then: shorten Explore round 3 to the circle only; Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:40 of 2:00

0:00	SLIDE	Show slide **Picture Equations** (s-title)
0:15	DO	Greet the whole group warmly, before the recording starts. Check that you can see each child, or ask the grown-up. [0:12]
0:27	SAY	Hello, inquirers! Pencil ready, and your shape cards, counters and two cups on the table. Cameras can be on or off, as you like.
0:39	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
0:56	DO	Start the recording now. Do not use children's first names on the recording. [0:05]
1:02	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook, shape cards, counters and two cups. Pause or rewind any time.*
1:14	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera.
1:28	SAY	Today a shape is a hidden number. The same shape is always the same number. We find it, then we check it.
	NOTE	Say the recording statement and the microphone routine in the same words every lesson, for anyone who joined late. Say the notice before you start the recording, and do not use children's first names once it is running. The routine and the four inquirer rules were introduced earlier in the term: do not re-teach them.
	NOTE	Plain statement for families when it comes up: a box with a hidden number in one line is Primary 1 work, but using the same shape in two lines and taking one line from another go beyond Primary 1 on purpose, always with counters and shape cards.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. The first recording of the term shows the full routine.

2:00 – 8:00 Remember warm-up

script 4:10 of 6:00

2:00	SLIDE	Show slide **Warm-up: a box hides a number** (s-rem1)
2:25	REC	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*
2:34	SAY	A box can hide a number. Look at the ten-frame: six counters, and the frame holds ten.
2:42	ASK	How many more counters fill the frame? Six plus box equals ten. [wait about 12 s for answers]
3:00	ASK	Box plus three equals eight. What is in the box? [wait about 12 s for answers]
3:17	ASK	Nine equals five plus box. What is in the box? [wait about 12 s for answers]
3:34	SLIDE	Show slide **The same number twice** (s-rem2)
3:59	SAY	Two plates with the same number of counters. Twelve counters in all.
4:05	ASK	What is five plus five? [wait about 12 s for answers]
4:20	ASK	Box plus box equals eight, the same number twice. What is in the box? [wait about 12 s for answers]
4:38	ASK	Seven plus seven? [wait about 12 s for answers]
4:52	SLIDE	Show slide **4 groups of 5** (s-rem3)

5:17	SAY	Four bags with five counters in each bag. Four times five means four groups of five.
5:25	ASK	How many counters are in four bags of five? [wait about 12 s for answers]
5:42	ASK	Three groups of four? [wait about 12 s for answers]
5:56	ASK	Two groups of ten? [wait about 12 s for answers]
	IF STUCK	If a child is unsure about the box, touch the empty places in the ten-frame and count them. For the same number twice, deal counters one at a time onto two plates. For the bags, add one bag at a time and say the total each time.
	NOTE	Answers: 4, 5, 4; 10, 4, 14; 20, 12, 20. Children answer with a whisper, fingers or hands. Nobody writes. Do not say picture equation yet. Joining cold: all you need is counting to 20 and adding small numbers.

8:00 – 12:00 Sneak-peek puzzles

script 3:04 of 4:00

8:00	SLIDE	Show slide **Sneak-peek puzzles** (s-sneak)
8:25	REC	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:34	SAY	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:47	READ	Puzzle one. A cup hides some counters. The cup and 4 loose counters make 9 counters in all. How many counters are in the cup? You may use counters.
9:03	CHILDREN	Children count out 9 counters or draw, and write the number on their own paper. [0:40]
9:43	READ	Puzzle two. Each shape is a number. Two stars make 14. The heart and a star make 10. How many is the heart? Write it.
9:56	CHILDREN	Children write or draw their answer. [1:00]
10:56	SAY	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: 5. Puzzle 2 answer: the star is 7, so the heart is 3. Look for good ideas to praise later (a child who found the star first). Do not write scores or keep a list.

12:00 – 24:00 Explore

script 9:44 of 12:00

12:00	SLIDE	Show slide **Explore: a shape and a pile** (s-explore1)
12:25	REC	*Recording: pause now. Use counters and your shape cards to find each hidden number. Press play when you have tried.*
12:35	SAY	Your hands are the teachers. Find your circle card and your square card. Count out ten counters. The picture says: circle plus square equals ten. The square is four.
12:50	CHILDREN	Children count out 10 counters, put 4 on the square card and the rest on the circle card. [1:40]
14:30	ASK	How many counters are on the circle card? [wait about 15 s for answers]
14:48	SAY	The circle is six. We check: six plus four is ten. It matches.
14:55	SLIDE	Show slide **The same shape twice** (s-explore2)
15:20	SAY	Round two. Heart plus heart equals twelve. Two hearts, so the same number on each.
15:28	CHILDREN	Children count out 12 counters and deal them one at a time onto two heart cards until the piles match. [1:40]
17:08	ASK	How many counters are on one heart card? [wait about 15 s for answers]
17:26	SAY	Both hearts are six. The same shape is always the same number.
17:32	ASK	Check it. Put the two piles together. How many counters? [wait about 15 s for answers]
17:52	SLIDE	Show slide **Two lines, same shapes** (s-explore3)

18:18	**SAY**	Round three. Two lines. Circle plus circle equals six. Circle plus triangle equals nine. Use line one to find the circle.
18:28	**CHILDREN**	Children find the circle from line one, then use the same circle in line two to find the triangle. Counters are fine. [2:10]
20:38	**ASK**	What number is the circle? [wait about 15 s for answers]
20:56	**ASK**	What number is the triangle? [wait about 15 s for answers]
21:13	**SAY**	The circle was three in both lines. That is our idea for today.
	IF STUCK	A child who puts a different number for the circle in line 2: ask, is it the same shape? A child who adds all the numbers in a line: ask, what is the total of the line? A child with no shape cards uses a cup or a piece of paper marked with the shape.
	NOTE	Main room only; children say their answers aloud and show fingers; you cannot see counters on small tiles. Answers: circle 6; heart 6 and 12 together; circle 3, triangle 6. Accept any child who finds the numbers by counters or by adding. Do not say equation for every line, say line.
21:20	**SLIDE**	Show slide **Two lines, same shapes** (s-explore3-ans)

24:00 – 36:00 Mini-lesson

script 9:57 of 12:00

24:00	**SLIDE**	Show slide **Three rules** (s-m1)
24:25	**REC**	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
24:32	**SAY**	Three rules for picture equations. One: each shape is a number. Two: the same shape is the same number every time. Three: in our puzzles, different shapes are different numbers.
24:46	**ASK**	Can a circle be six in one line and four in another line of the same puzzle? Thumb up or down. [wait about 20 s for answers]
25:17	**SLIDE**	Show slide **Find it, then check it** (s-m2)
25:42	**SAY**	Find it, then check it. Circle plus circle plus square equals sixteen. The square is four.
25:50	**ASK**	Take the square away from sixteen. What is left for the two circles? [wait about 15 s for answers]
26:12	**ASK**	Two equal circles make twelve. What is one circle? [wait about 15 s for answers]
26:31	**SAY**	Now we check. Six plus six plus four equals sixteen. It matches.
26:37	**SLIDE**	Show slide **Find it, then check it** (s-m2-ans)
27:02	**SLIDE**	Show slide **Take one line from another** (s-m3)
27:27	**SAY**	Two lines at once. Star plus star plus triangle equals seventeen. Star plus triangle equals twelve. Line one has one more star than line two.
27:40	**ASK**	How much more is seventeen than twelve? [wait about 15 s for answers]
27:58	**SAY**	So that one extra star is five.
28:02	**ASK**	Five plus the triangle makes twelve. What is the triangle? [wait about 15 s for answers]
28:22	**SLIDE**	Show slide **Take one line from another** (s-m3-ans)
28:46	**SLIDE**	Show slide **Check in both lines** (s-m3b)
29:12	**SAY**	We always check in both lines. Five plus five plus seven is seventeen. Five plus seven is twelve. Both lines match.
29:22	**SLIDE**	Show slide **A times line** (s-m4)
29:47	**SAY**	A times line. Heart times circle equals twelve. The circle is three. That means heart groups of three make twelve.
29:57	**ASK**	How many groups of three make twelve? [wait about 20 s for answers]
30:20	**SAY**	The heart is four. Check: four times three is twelve.

30:26	**SLIDE**	Show slide **Your turn** (s-m5)
30:50	**READ**	Your turn. Star plus heart equals eleven. Star plus star plus heart equals sixteen. Find the star and the heart. Then check both lines.
31:04	**CHILDREN**	Children work on their own paper, with counters if they like. Wait 10 seconds before helping. [2:20]
33:24	**SLIDE**	Show slide **Check: your turn** (s-m5a)
33:49	**READ**	The star is five. The heart is six. Tick your own work with a tick.
	NOTE	Key words to use every time: shape, line, the same number, check. Say take one line from the other. Never use letters. Beyond Primary 1 on purpose: using the same shape in two lines, taking one line from another and the times line. Do not teach a rule to recite; the counters are the method.
	IF STUCK	If a child adds every number they see, ask: which numbers does the line really give us? If a child cannot start with two lines, make the second line smaller with counters: 11 counters, then 16 counters, and ask what is extra.

36:00 – 40:00 Brain break

script 3:17 of 4:00

36:00	**SLIDE**	Show slide **Clap and whisper** (s-break)
36:25	**REC**	*Recording: stand up and join in! Count to 20, clap on 5, 10, 15 and 20, and whisper the rest.*
36:35	**SAY**	Stand up and get a sip of water. I count from one. Whisper every number, and clap loudly on five, ten, fifteen and twenty.
36:47	**CHILDREN**	Count 1 to 20 together: whisper the numbers and clap on 5, 10, 15 and 20. Then count back from 20 in fives. Children may stay seated. [2:30]

40:00 – 70:00 Workbook

script 24:58 of 30:00

40:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
40:25	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
40:34	**READ**	Page 1, Start here. Question 1: a cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. How many counters are in the cup? Question 2: each shape is a number. Circle plus square equals 10. The square is 4. Find the number for the circle. Question 3: heart plus heart equals 16. Find the number for the heart. Question 4: star plus star plus star equals 15. Find the number for the star.
41:18	**CHILDREN**	Page 1, questions 1 to 4. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [8:50]
50:08	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
50:33	**READ**	Question 1: 8. Question 2: 6. Question 3: 8. Question 4: 5. Tick your own page with a tick.
50:43	**SAY**	After question 4 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
50:52	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
51:17	**READ**	Page 2. Question 5: two triangles and a square make 14. The square is 4. Fill in the blanks. Question 6: circle plus circle equals 12. Ben says the circle is 7. Is Ben correct? Circle Yes or No, then fill in the circle. Question 7: each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all.
51:53	**CHILDREN**	Page 2, questions 5 to 7. [8:50]
60:43	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
61:08	**READ**	Question 5: 10 and 5. Question 6: no, the circle is 6. Question 7: 10 and 5.
61:17	**SLIDE**	Show slide **Choose your next page** (s-choice)
61:42	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.

61:58	**MOVE**	Stand and stretch for 30 seconds, then sit down and choose.
62:28	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [2:30]
	IF STUCK	Q1: take the loose counters away from 11. Q2: take the square away from the total. Q3: share the total between two hearts. Q4: share 15 between three stars. Q5: take the square away, then share. Q6: put 7 into both circles and add. Q7: take the loose sweets away, then share between two bags.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder uses two lines and a times line (beyond Primary 1 on purpose). Go further has three shapes and a sum and a product. Do not announce who is on which page. Start here takes a middle child about 18 minutes, so the choice time is short: for most children pages 3 to 5 are for home. Say so plainly, do not skip the answer slides, and take any extra minutes from the flex time.

70:00 – 76:00 Talk time

script 4:50 of 6:00

70:00	**SLIDE**	Show slide **Talk and explain** (s-talk)
70:25	**REC**	*Recording: pause. Tell a grown-up if Aisha is right, and test it with counters.*
70:32	**READ**	Aisha says: circle plus circle equals twelve, so circle plus circle plus circle equals eighteen. Is Aisha right? How could we check?
70:44	**CHILDREN**	Test Aisha with counters: 12 on two circle cards, then three circle cards. [1:00]
71:44	**CHILDREN**	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]
72:24	**ASK**	Who would like to share? Volunteers first. A pass is always fine. [wait about 80 s for answers]
73:50	**SLIDE**	Show slide **Back to the sneak-peek** (s-sneakans)
74:15	**SAY**	Remember the sneak-peek? Two stars make fourteen, so the star is seven. The heart and seven make ten, so the heart is three. The cup held five.
74:28	**ASK**	Did anyone change their mind while we worked? You can just nod. [wait about 15 s for answers]
	NOTE	Model answer: yes. The same shape is the same number, so each circle is 6 (12 shared between two), and three circles are 3 sixes: 18. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:53 of 6:00

76:00	**SLIDE**	Show slide **Exit ticket** (s-exit)
76:25	**REC**	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	**SAY**	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:41	**READ**	E1. Each shape is a number. The triangle plus the triangle equals 18. Find the number for the triangle. E2. The heart plus the square equals 13. The square is 5. Find the number for the heart. Choose the correct answer. Option 1: 5. Option 2: 6. Option 3: 7. Option 4: 8. Write its number in the bracket. E3. Each shape is a number. Circle plus circle plus star equals 19. The star is 5. Find the number for each circle.
77:26	**CHILDREN**	Children answer E1 to E3 on their own. [2:50]
80:16	**SLIDE**	Show slide **Check the exit ticket** (s-exitans)
80:41	**READ**	E1: 9. E2: option 4, the heart is 8. E3: each circle is 7.
80:48	**SAY**	Tick your own ticket with a tick. Well done.

	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result.

82:00 – 85:00 Look-back

script 2:23 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:25	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
83:03	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:57 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:25	**SAY**	Well done, inquirers. Next lesson is Find One First, and it holds Mi's Challenge Quest, a game with Mi. Please have your Term 2 Lesson 10 workbook with its tokens page cut out, three to six small things for a shop, your Quest Log in its envelope (keep it closed), a quest partner, a pencil and paper. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: one line or two?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: circle plus square equals eight. Could the circle be two? Could it be three? What does the second line tell us?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: line 1 alone has many answers; line 2 makes only circle 3 and square 5 fit both. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask ‘How do you know?’ and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer

Can a shape change?	4 (optional)	Ali says: the heart is 5 in line 1 and 6 in line 2, and both lines are fine. Is that allowed? Test it with counters: put the same heart card into both lines.	*Recording: pause. Ali says the heart is 5 in one line and 6 in another. Is that allowed? Test it with counters.*	No. The same shape is the same number in every line. If the heart is 5 in line 1 it must be 5 in line 2 too. If the numbers do not fit, one of them is wrong, so check again.
Make a puzzle for your grown-up	4 (optional)	Pick a number for the circle and a different number for the star. Write two lines, such as circle + star = a total and circle + circle + star = a total. Ask your grown-up to find your numbers.	*Recording: pause. Pick two numbers. Write two lines for a grown-up to solve, then check their answer.*	No single answer. Check the totals: with circle 4 and star 6, line 1 is 10 and line 2 is 14. The grown-up takes one line from the other.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never met a picture equation. Use a cup and counters to show how a shape can hide a number.	*Recording: pause. Show a grown-up how a cup can hide a number and how to find it.*	A cup and 4 loose counters make 9 in all. Count out 9, take the 4 away, and the rest is what the cup holds: 5.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): A cup hides some counters. The cup and 4 loose counters make 9 counters in all. How many counters are in the cup?

Answer: 5

Puzzle 2 (think): Each shape is a number. The star + the star = 14. The heart + the star = 10. Find the number for the heart. Write it.

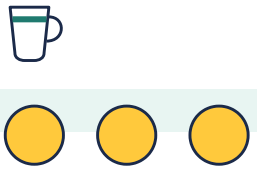
Answer: 3. The two stars make 14, so one star is 7. The heart and 7 make 10, so the heart is 3.

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **A cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. Fill in the blank.**

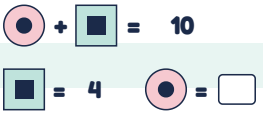
The cup holds ☐ counters.



Answer: 8

Hint (say this first): Take the 3 loose counters away from 11.
Worked solution: 11 - 3 = 8, so the cup holds 8 counters. Check: 8 + 3 = 11.
Common mistakes: Writing 14 (11 + 3).; Writing 11 (the total).
Time for a middle child: about 2 min.

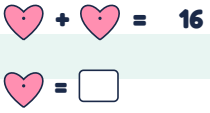
S2 **Each shape is a number. Find the number for the circle.**



Answer: 6

Hint (say this first): Take the square away from the total: 10 - 4.
Worked solution: The total is 10 and the square is 4. 10 - 4 = 6, so the circle is 6. Check: 6 + 4 = 10.
Common mistakes: Writing 14 (adding 10 and 4).; Writing 10 (the total).
Time for a middle child: about 2 min.

S3 **Each shape is a number. Find the number for the heart.**



Answer: 8

Hint (say this first): Two hearts are the same number. Share 16 between the two hearts.
Worked solution: 8 + 8 = 16, so each heart is 8.
Common mistakes: Writing 16 (the total).; Writing 32 (adding the total twice).
Time for a middle child: about 2 min.

S4 **Each shape is a number. Find the number for the star.**



Answer: 5

Hint (say this first): The 3 stars are the same number. Share 15 between 3 stars.
Worked solution: 5 + 5 + 5 = 15, so each star is 5.
Common mistakes: Writing 3 (counting the stars).; Writing 12 (15 - 3).
Time for a middle child: about 2 min.

S5 **Each shape is a number. The square is 4. Fill in the blanks.**



(a) The 2 triangles make ☐☐.
(b) Each triangle is ☐.

Answer: 10; 5

Hint (say this first): Take the square away from 14 first.
Worked solution: 14 - 4 = 10, so the 2 triangles make 10. The triangles are the same, so each is 5. Check: 5 + 5 + 4 = 14.
Common mistakes: Writing 14 for each triangle.; Writing 10 for each triangle.
Time for a middle child: about 3 min.

S6 **Ben says the circle is 7. Look at the picture.**



(a) Is Ben correct? Circle Yes or No.
(b) The circle is ☐.

Answer: No; 6

Hint (say this first): Put 7 into both circles. What is the total?
Worked solution: 7 + 7 = 14, but the line says 12. So Ben is not correct. 6 + 6 = 12, so the circle is 6.
Common mistakes: Saying Yes because 7 is close to 6.; Writing 12 for the circle.
Time for a middle child: about 3 min.

S7 **Each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all. Fill in the blanks.**



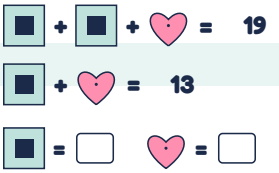
(a) The 2 bags have ☐☐ sweets in all.
(b) Each bag has ☐ sweets.

Answer: 10; 5

Hint (say this first): Take the 3 loose sweets away from 13.
Worked solution: 13 - 3 = 10 sweets are in the 2 bags. 5 + 5 = 10, so each bag has 5 sweets. Check: 5 + 5 + 3 = 13.
Common mistakes: Writing 13 for each bag.; Writing 6 (13 shared between 2 bags, ignoring the loose sweets).
Time for a middle child: about 3.5 min.

Think harder (page 3): the taught idea in a new setting

T1 **Each shape is a number. Find the number for each shape. Show your working.**



Answer: square 6; heart 7

Hint (say this first): Line 1 has one more square than line 2. How much more is 19 than 13?
Worked solution: Line 1 has one more square than line 2, and 19 is 6 more than 13, so the square is 6. Then 6 + heart = 13, so the heart is 7. Check: 6 + 6 + 7 = 19.
Common mistakes: Writing 19 and 13 as the answers.; Finding the square but forgetting the heart.
Time for a middle child: about 4 min.

True/False statements: Line 1 is true with Kumar's numbers.; Line 2 is true with Kumar's numbers.

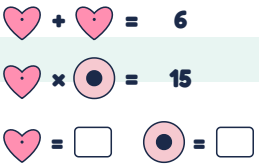
T2 **Kumar says the heart is 8 and the star is 7. Look at the picture. Decide if each statement is True or False.**



Answer: True; False

Hint (say this first): Put 8 and 7 into each line and add.
Worked solution: Line 1: 8 + 7 = 15, which matches. Line 2: 8 + 8 + 7 = 23, but the line says 21. Kumar's numbers work in line 1 only.
Common mistakes: Checking only line 1.; Saying line 2 is true because 23 is close to 21.
Time for a middle child: about 3 min.

T3 **Each shape is a number. Find the number for each shape.**

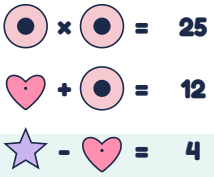


Answer: heart 3; circle 5

Hint (say this first): Line 1 gives the heart. Then use the heart in line 2.
Worked solution: The 2 hearts make 6, so the heart is 3. Then 3 × circle = 15 means 3 groups of 5 make 15, so the circle is 5. Check: 3 × 5 = 15.
Common mistakes: Writing 6 for the heart.; Writing 15 for the circle.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **Each shape is a number. Find the number for each shape.**



(a) The circle is ☐.
(b) The heart is ☐.
(c) The star is ☐.

Answer: circle 5; heart 7; star 11

Hint (say this first): Start with the line that has only one kind of shape.
Worked solution: Circle × circle = 25, so the circle is 5 (5 groups of 5). Then heart + 5 = 12, so the heart is 7. Then star - 7 = 4, so the star is 11. Check: 11 - 7 = 4.
Common mistakes: Writing 12 for the heart.; Taking 4 away from 7 and writing 3 for the star.
Time for a middle child: about 5 min.

G2 **Each shape is a number. Fill in the blanks.**



(a) One star and one heart make ☐☐.
(b) The star is ☐.
(c) The heart is ☐.

Answer: 14; star 8; heart 6

Hint (say this first): Put the two lines together. How many stars and hearts are there?
Worked solution: Both lines together have 3 stars and 3 hearts, and 22 + 20 = 42. 3 stars and 3 hearts make 42, so one star and one heart make 14. Then 22 - 14 = 8 is a star, and 14 - 8 = 6 is a heart.
Common mistakes: Writing 22 and 20 as the answers.; Writing 7 for both shapes.
Time for a middle child: about 5 min.

G3 **Each shape is a number. The heart is greater than the triangle. Find the number for each shape.**



(a) The heart is ☐.
(b) The triangle is ☐.

Answer: heart 5; triangle 4

Hint (say this first): Which two numbers add to 9? Try pairs and multiply each pair.
Worked solution: The pairs that add to 9 are 1 and 8, 2 and 7, 3 and 6, 4 and 5. Only 4 and 5 multiply to 20. The heart is greater, so the heart is 5 and the triangle is 4.
Common mistakes: Writing 4 for the heart.; Using 3 and 6 (they add to 9 but multiply to 18).
Time for a middle child: about 5 min.

Investigation (page 5): One line or two?

Easy first step: The circle + the triangle = 10. One pair: circle 2, triangle 8. Write 2 more.
Then: The circle + the circle + the triangle = 14. Do any pairs fit both lines? Why?

Model answer: Only circle 4 and triangle 6 fit both lines. One line alone has many pairs that fit (1 and 9, 2 and 8, 3 and 7, 4 and 6, 5 and 5 and more; accept 5 and 5 for line 1). 5 and 5 fails line 2 (5 + 5 + 5 = 15, not 14). The second line is the same as the first with one more circle, and 14 is 4 more than 10, so the circle is 4.

Look for: A child who tries pairs for both lines, finds that only one pair works, or says one line is not enough to find two numbers. Credit any clear reason. Do not mark a child down for stopping at one pair.
Common mistakes: Choosing a pair that fits line 1 only, such as 5 and 5 (it makes 15 in line 2).; Stopping after line 1.

Talk and explain (minutes 70 to 76)

Aisha says: circle + circle = 12, so circle + circle + circle = 18. Is Aisha right? How could we check? Sentence starter: I think ... because ...

> **Model answer: Yes. The same shape is the same number, so each circle is 6 (12 shared between two). Three circles are 6 + 6 + 6 = 18. We can check with counters: 12 on two cards is 6 on each, then add a third card of 6.**

Look for: A child who finds the circle first, or who uses counters or a picture. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Each shape is a number. The triangle + the triangle = 18. Find the number for the triangle.**

The triangle is ▢.

> **Answer: 9**

Worked solution: 9 + 9 = 18, so each triangle is 9.
Common mistakes: Writing 18 (the total).; Writing 2 (counting the triangles).

E2 **The heart + the square = 13. The square is 5. Find the number for the heart. Choose the correct answer. Write its number in the bracket.**

> **Answer: (4) 8**

Worked solution: 13 - 5 = 8, so the heart is 8. Check: 8 + 5 = 13.
Common mistakes: Choosing (1) 5, the number of the square.; Choosing (2) 6 or (3) 7 after a counting slip.

E3 **Each shape is a number. The star is 5. Find the number for each circle.**



Each circle is ▢.

> **Answer: 7**

Worked solution: 19 - 5 = 14, so the 2 circles make 14. 7 + 7 = 14, so each circle is 7. Check: 7 + 7 + 5 = 19.
Common mistakes: Writing 14 for each circle.; Writing 19 for each circle.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (finding the number for one shape; using two lines; checking in every line; something else); *What helped me* (counters and shape cards; drawing it; checking my answer; I was not stuck); *Next time I will try* (using counters; checking my answer; drawing it; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Each shape is a number. The star + the star = 8. Find the number for the star. ___	4
H2	Start here	A cup hides some counters. The cup and 5 loose counters make 14 counters in all. How many counters are in the cup? ___	9
H3	Start here	The heart is 4. What is the heart + the heart + the heart? ___	12
H4	Think harder	Line 1: the circle + the circle + the triangle = 15. Line 2: the circle + the triangle = 9. Find the number for the triangle. ___	3
H5	Go further	Line 1: the circle × the circle = 16. Line 2: the heart + the circle = 9. (× means groups of.) Find the number for the heart. ___	5

Online practice (`out/l19/practice`, key `t2l19`): 12 items; 11 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no

scores, no streaks.

6. Parent line for the term report

Your child is learning that a shape can stand for a hidden number and that the same shape is always the same number. This week: finding one shape, then two shapes from two lines.

Term line (plan, section 5): This term: strips and stories so far; now picture equations; next, prices and Mi's Challenge Quest. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Picture Equations** s-title	TODAY WE WILL: / Find the number a shape is hiding.	Term 2 Lesson 9, 90 minutes. Objective: find a hidden number behind a shape in a picture equation, know that the same shape is always the same number, check the answer in every line, and take one line away from another. Plain statement for families: one line with a hidden number is Primary 1 work, but using the same shape in two lines, taking one line from another and a times line go past the school Primary 1 syllabus on purpose. Shapes and boxes only, never letters. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 24, mini-lesson 24 to 36, brain break 36 to 40, workbook 40 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Say the recording notice in one line: you and the screen only; a child may keep the camera off.
2	2–8	**Warm-up: a box hides a number** s-rem1	Touch the counters. Say the number in the box. / [picture: ten_frame] / [picture: table]	Six counters sit in a ten-frame that holds ten. The box hides how many more fill it. Ask: how many more fill the frame (4); the box in the second line (5); the box in the third line (4). Children answer aloud or with fingers. No writing, no marks. A child who is unsure touches each counter, counts the empty places (children may lay counters on a ten-frame of the kit page), or holds up fingers for the first number and counts on.
3	2–8	**The same number twice** s-rem2	[picture: groups] / [picture: table] / Two plates with the same number of counters.	Two plates with the same number of counters. Twelve counters in all. Ask: what is 5 + 5 (10); box + box = 8, the same number twice (4); 7 + 7 (14). Children answer aloud. A child who is unsure deals counters one at a time onto two plates, or uses fingers on two hands. The same number twice is a double: it comes back with the hearts in Explore.
4	2–8	**4 groups of 5** s-rem3	[picture: groups] / [picture: table] / 4 × 5 means 4 groups of 5.	Four bags with five counters in each bag. We add one bag at a time: 5, 10, 15, 20. Ask: how many counters in 4 bags of 5 (20); 3 groups of 4 (12); 2 groups of 10 (20). This is Term 2 Lesson 5 taught again: the × sign says groups of. A child who is unsure counts the counters one bag at a time, saying the total each time.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / A cup hides some counters. / The cup and 4 loose counters make 9 in all. / How many counters are in the cup? / PUZZLE 2: think: / Each shape is a number. / How many is the heart? / [picture: picture_equation]	Puzzle 1 (a cup and 4 counters make 9; the cup holds 5) is an access question: any child can count out 9 and take 4 away. Puzzle 2 is a thinking question on the lesson idea: two stars make 14, so a star is 7, and the heart and a star make 10, so the heart is 3. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: 5; 3. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.
6	12–24	**Explore: a shape and a pile** s-explore1	[picture: picture_equation] / Count out 10 counters. / Put 4 on the square card. / The rest go on the circle card. How many?	Round 1, about 5 minutes. Children use their shape cards (the cut-out kit cards: circle, square, triangle, heart and star, up to three of each) and counters (the kit page counters or small things). The picture says circle + square = 10, and the square is 4. They count out 10 counters, put 4 on the square card and the rest on the circle card, then count the circle card (6). Say: the circle is 6. Write circle = 6 on the screen, then check: 6 + 4 = 10. If a family has no cut-out cards, a cup or a piece of paper with the shape drawn on it works. Answer: 6. Do not name a child with a wrong answer.

7	12–24	**The same shape twice** s-explore2	[picture: picture_equation] / Count out 12 counters. / Share them onto 2 heart cards, one at a time. / The piles must match. How many on one heart?	Round 2, about 5 minutes. Heart + heart = 12. Both hearts are the same number, so the piles must match. Children deal 12 counters one at a time onto two heart cards. Ask: how many counters are on one heart card (6). Say: both hearts are 6, because the same shape is always the same number. Then check: put the piles together and count (12). Answer: 6. A child who deals unevenly is asked: are the piles the same?
8	12–24	**Two lines, same shapes** s-explore3	[picture: picture_equation]	Round 3, about 4 minutes. Two lines: circle + circle = 6 and circle + triangle = 9. Children use the first line to find the circle (3), then the same circle in the second line to find the triangle (6). They may use counters: 6 on two circle cards, then 9 on one circle card and one triangle card. Ask: what number is the circle (3), the triangle (6). Say: the circle is 3 in both lines. The two lines on the slide are shown first; the two sentences about the circle appear after the children have tried (the next slide). Answers: circle 3, triangle 6. A child who gets a different circle in line 2 is asked: is it the same shape? The numbers are not the sneak-peek numbers: do not tell the sneak-peek answers here.
9	12–24	**Two lines, same shapes** s-explore3-ans	[picture: picture_equation] / Line 1 tells you the circle. / The circle is the same number in line 2.	Round 3, about 4 minutes. Two lines: circle + circle = 6 and circle + triangle = 9. Children use the first line to find the circle (3), then the same circle in the second line to find the triangle (6). They may use counters: 6 on two circle cards, then 9 on one circle card and one triangle card. Ask: what number is the circle (3), the triangle (6). Say: the circle is 3 in both lines. The two lines on the slide are shown first; the two sentences about the circle appear after the children have tried (the next slide). Answers: circle 3, triangle 6. A child who gets a different circle in line 2 is asked: is it the same shape? The numbers are not the sneak-peek numbers: do not tell the sneak-peek answers here. (answer shown after the children have answered)
10	24–36	**Three rules** s-m1	Each shape is a number. / The same shape is the same number every time. / In our puzzles, different shapes are different numbers.	Say the three rules once, slowly. Rule 3 is a rule of our puzzles only (in other maths two shapes can be the same number), so that each puzzle has one answer. Ask: can a circle be 6 in one line and 4 in another line of the same puzzle (no, rule 2). Children answer with a thumb. Never use letters; the shapes are the unknown numbers.
11	24–36	**Find it, then check it** s-m2	[picture: picture_equation]	Circle + circle + square = 16 and the square is 4. Take the square away: 16 - 4 = 12 for the two circles. The two circles are equal, so each is 6. Check by putting 6 back into the line: 6 + 6 + 4 = 16. The steps card appears after the questions (next slide); children say each number first. Ask: what is left for the two circles (12); what is one circle (6). Always end with the check.
12	24–36	**Find it, then check it** s-m2-ans	[picture: picture_equation] / STEPS: / Take the square away from the total. / Share what is left between the 2 equal circles. / Check: put the number back into the line.	Circle + circle + square = 16 and the square is 4. Take the square away: 16 - 4 = 12 for the two circles. The two circles are equal, so each is 6. Check by putting 6 back into the line: 6 + 6 + 4 = 16. The steps card appears after the questions (next slide); children say each number first. Ask: what is left for the two circles (12); what is one circle (6). Always end with the check. (answer shown after the children have answered)
13	24–36	**Take one line from another** s-m3	[picture: picture_equation]	Two lines about the same shapes: star + star + triangle = 17 and star + triangle = 12. Line 1 has one more star than line 2, and 17 is 5 more than 12, so one star is 5. Then star + triangle = 12, so 5 + triangle = 12 and the triangle is 7. Ask: how much more is 17 than 12 (5); what is the triangle (7). The steps appear on the next slide, after the children have answered. This is the new idea of the lesson and goes beyond Primary 1: say it was found by taking one line from the other.

14	24–36	**Take one line from another** s-m3-ans	[picture: picture_equation] / STEPS: / Line 1 has one more star than line 2. / The totals show how much that extra star is. / Then put the star into line 2 to find the triangle.	Two lines about the same shapes: star + star + triangle = 17 and star + triangle = 12. Line 1 has one more star than line 2, and 17 is 5 more than 12, so one star is 5. Then star + triangle = 12, so 5 + triangle = 12 and the triangle is 7. Ask: how much more is 17 than 12 (5); what is the triangle (7). The steps appear on the next slide, after the children have answered. This is the new idea of the lesson and goes beyond Primary 1: say it was found by taking one line from the other. (answer shown after the children have answered)
15	24–36	**Check in both lines** s-m3b	Line | Check | Does it match? / Line 1 | 5 + 5 + 7 = 17 | Yes / Line 2 | 5 + 7 = 12 | Yes / Check in every line, not just one.	A number that works in one line may not work in another, so we check every line. Line 1: 5 + 5 + 7 = 17. Line 2: 5 + 7 = 12. Both match. Ask: what would we think if line 1 did not match (the numbers are wrong, try again). Link to the Think harder question where a friend gives numbers that work in only one line.
16	24–36	**A times line** s-m4	[picture: picture_equation] / STEPS: / heart × circle means heart groups of the circle. / How many groups of the circle make the total? / Check: put the number back into the line.	A times line: heart × circle = 12 and the circle is 3. Read it as heart groups of 3 make 12. Ask: how many groups of 3 make 12 (4). Children may count in 3s on fingers or put counters in groups of 3. Check: 4 × 3 = 12. Either reading order is accepted in a child answer. Times lines come back in Think harder and Go further.
17	24–36	**Your turn** s-m5	[picture: picture_equation] / Find the star and the heart. Then check both lines.	Each child works on their own paper with counters if they like. Star + heart = 11 and star + star + heart = 16. Line 2 has one more star than line 1, and 16 is 5 more than 11, so the star is 5 and the heart is 6. Walk the cameras; do not ask children to hold papers up. Wait 10 seconds before helping. Answers: star 5, heart 6.
18	24–36	**Check: your turn** s-m5a	Shape | Number | Check / Star | 5 | Line 1: 5 + 6 = 11 / Heart | 6 | Line 2: 5 + 5 + 6 = 16	Answers: star 5, heart 6. Read each answer aloud so a child watching the recording can mark their own work. Children tick their own work with a tick. Never ask a child to show a wrong answer to the group.
19	36–40	**Clap and whisper** s-break	Count to 20 with me. Clap on 5, 10, 15 and 20. / Whisper the numbers in between. / Stand up. Sip some water.	Four minutes. Stand up on camera. Count from 1 to 20 together: children whisper every number and clap loudly on 5, 10, 15 and 20. Then count back from 20 in fives if there is time. No equipment, no running, a little clear space. A child may stay seated. Do not give prizes or compare children.
20	40–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 4. Take your time. / questions S1, S2, S3, S4	Read questions 1 to 4 aloud. Children work on their own with counters and shape cards; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 4 any child may choose to carry on or try Think harder. Question 1: a cup hides counters; 11 in all and 3 are loose. Question 2: circle + square = 10 and the square is 4. Question 3: heart + heart = 16. Question 4: three stars make 15.
21	40–70	**Check page 1** s-ws1ans	answers S1, S2, S3, S4	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1 8; Q2 6; Q3 8; Q4 5.
22	40–70	**Workbook page 2: Start here** s-ws2	Questions 5 to 7. Take your time. / questions S5, S6, S7	Read questions 5 to 7 aloud. Question 5: two triangles and a square make 14, and the square is 4. Question 6: Ben says the circle is 7; circle + circle = 12. Question 7: 2 bags and 3 loose sweets make 13.
23	40–70	**Check page 2** s-ws2ans	answers S5, S6, S7	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q5 10 and 5; Q6 No, the circle is 6; Q7 10 and 5.

24	40–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Two lines and a times line / Go further (page 4) | Three shapes at once / Investigation (page 5) | One line or two? / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
25	70–76	**Talk and explain** s-talk	AISHA SAYS: / Circle + circle = 12. / So circle + circle + circle = 18. / [picture: picture_equation] / Is Aisha right? How could we check? / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Children test Aisha with counters: 12 counters on two circle cards is 6 each, and three circle cards of 6 make 18. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: yes. The same shape is the same number, so each circle is 6, and three sixes are 18. Repeat each child answer in your own words for the recording.
26	70–76	**Back to the sneak-peek** s-sneakans	[picture: picture_equation] / The star is 7. The heart is 3. The cup holds 5.	Reveal both puzzles now. Say: many of you wrote a good reason. Do not read out anyone earlier answer. Two stars make 14, so one star is 7; the heart and 7 make 10, so the heart is 3. Puzzle 1: 9 counters in all and 4 loose, so the cup holds 5. It was the same idea: a hidden number.
27	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud with their totals, and the four options of E2 (the figures on the slide are small). Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
28	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: triangle 9; option 4, the heart is 8; each circle is 7.
29	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.
30	85–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out today. / NEXT LESSON: / Find One First, and Mi's Challenge Quest / Have ready: the Term 2 Lesson 10 workbook with its tokens page cut out (30 tokens), 3 to 6 small things, the Quest Log in its envelope (keep it closed), a quest partner, a pencil and paper.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that next lesson holds Mi's Challenge Quest, a game with Mi: the Quest Log and the grown-up note were sent a week before. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide.
31	optional	**Go further: one line or two?** s-opt	[picture: picture_equation] / Could the circle be 2? Could it be 3? / [picture: picture_equation] / What does the second line tell us?	Optional, only if you are on time. Line 1 alone has many answers: circle 2 and square 6, circle 3 and square 5, circle 1 and square 7 and more. Line 2 (circle + circle + square = 11) has only one pair that fits both: circle 3 and square 5. Check: 3 + 5 = 8 and 3 + 3 + 5 = 11. Take two answers and move on. This links to the Investigation.

32	optional	**If we have time: pick one** s-menu	Activity | What we do / Can a shape change? | Ali says the heart is 5 in one line and 6 in another. / Make a puzzle | Pick two numbers. Write two lines for your grown-up. / Teach it back | Show a toy how a cup hides a number / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook, shape cards, counters and two cups. Pause or rewind any time.*	Gather your workbook, shape cards, counters, two cups, pencil and paper. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*	Play along. Say each answer out loud before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Use counters and your shape cards to find each hidden number. Press play when you have tried.*	Pause. Put counters on the shape cards to find each hidden number, then check it.
24:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
36:00	Brain break	*Recording: stand up and join in! Count to 20, clap on 5, 10, 15 and 20, and whisper the rest.*	Stand up. Clap on 5, 10, 15 and 20 and whisper the numbers in between.
40:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up if Aisha is right, and test it with counters.*	Pause. Explain your answer to someone at home, and test it with counters.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:40	1:53	ok
remember	6:00	4:10	4:56	LOOSE
sneak	4:00	3:04	3:35	ok
explore	12:00	9:44	11:35	ok

mini	12:00	9:57	11:46	ok
break	4:00	3:17	3:54	ok
work	30:00	24:58	29:45	ok
talk	6:00	4:50	5:44	ok
exit	6:00	4:53	5:44	ok
look	3:00	2:23	2:48	ok
flex	5:00	0:57	1:05	ok
Total	90:00	69:52	82:46	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 7 questions; a middle child needs about 8.0 min for page 1 and 9.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.